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EVALUATION OF POLICIES, STATE-OF-THE-ART, AND FUTURE RESEARCH
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Evaluation of Policies, State-of-the-Art, and Future Research Directions in Database Security

- Extended Abstract -

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Abstract: We briefly describe the history of database security, define its scope, and by taking a critical point of view perform an analysis of different proposals made. Main emphasis is given to access control techniques and the discretionary access controls, the mandatory access controls, and the adapted mandatory access control (AMAC) model are discussed. After reviewing the current state-of-the-art we point out some directions in which we expect the entire field to move within the next few years.

Key words: Database Security, Access Controls, Research Directions, Security Policies.

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1. Introduction

The field of database security has been active for over twenty five years since census agencies in the United States, Canada, and Sweden began to process their statistical data by using data processing technology. During these early days the concept of data security became established when mathematicians and statisticians began to look at the statistical inference problem which arises whenever sensitive information can be inferred from statistical summaries of the census data. It was only shortly later when the relational

model was proposed as a mathematical basis for dealing with data and the research focus shifted to access control issues which specify the rules under which users can create, update, and delete data objects. The access matrix model was introduced and has been used to maintain authorizations. A milestone was the development of system R in the mid seventies which implemented a powerful representation of the matrix special-suited for relational systems, namely an authorization relation and discretionary access controls [1]. In the mid and late seventies there was a great deal of interest in the theoretical aspects of these models (for references, see [2]). Then most of the relational database security research turned to other security techniques. However, the appearance of more advanced data models has renewed interest in discretionary policies recently.

As databases became larger and larger and technology evolved, DBMSs were also chosen to process high sensitive information. As a consequence the need for having a security policy that addresses a higher level of threat than discretionary policies became evident. The turning point into the research came in the mid eighties when computer scientists began to apply mandatory protection techniques to databases. As a result multilevel data models were developed, and much of the research potential in science and industry has been devoted to handle and understand this additional complexity. There are still many open questions to solve but mandatory protection has already proven to offer very effective security controls. This is recognized by the national and international computer security evaluation criteria that require for systems in order to get evaluated at the higher classes of security (the US criteria, for example, for B1 and above) the support of mandatory access controls.

Pure mandatory protection were initially developed for the use in operating systems and it is the opinion of the authors that they look somewhat forced when applied to databases. A careful study of their limitations has led to the development of the Adapted Mandatory Access Control (AMAC) model, a technique special-suited for the security requirements of databases. We will discuss the different security policies in section 2 of this paper in a nutshell. For a more detailed study consult [3].

The scope of database security is very broad and entails such things as moral and ethical issues imposed by public and society, legal issues where control is legislated over the collection and disclosure of stored information, or more technical issues such as how to protect the stored information from loss or unauthorized access, destruction, use, modification, or disclosure. It is generally understood that database security must be concerned with ensuring the secrecy, integrity, and availability of data stored in a database. To define the terms, *secrecy* denotes the protection of information from unauthorized disclosure either by direct retrieval or by indirect logical inference. In addition, secrecy must deal with the possibility that information may also be disclosed by legitimated users acting as an 'information channel' by passing secret information to unauthorized users. This may be done intentionally or without knowledge of the authorized user. *Integrity* requires data to be protected from malicious or accidental modification, including the insertion of false data, the contamination of data, and the destruction of data. Integrity constraints are rules that define the correct states of a database and thus can protect the correctness of the database during operation. *Availability* is the characteristic that ensures data being available to authorized users when they need them. Availability includes the 'denial of service' of a system, i. e. a system is not operating in accordance with its intended purpose. Availability is closely related to integrity because 'denial of service' may be caused by unauthorized destruction, modification, or delay of service as well.

2. Database Security Policies Revisited

To ensure a certain degree of database security the computing environment must at least support identification and authentication, authorization and access controls, database integrity and consistency, and the auditing of all security relevant actions. In this paper such a broad perspective is not taken and main focus is directed towards aspects related to authorization and access controls. This is legitimate because identification, authentication,

and auditing¹ normally fall within the scope of the underlying operating system and integrity and consistency policies are subject to the closely related topic of 'semantic data modelling' or are dependent on the physical design of the DBMS software (namely, on the transaction and recovery manager).

2.1 Discretionary Access Controls

Discretionary access controls (DAC) are based on the concepts of a set of security objects O , a set of security subjects S , a set of access privileges T defining what kind of access a subject has to a certain object, and in order to represent content-based access rules a set of predicates P . Applied to databases O is a finite set of values $\{o_1, \dots, o_n\}$ representing data schemas, S is a finite set of potential subjects $\{s_1, \dots, s_m\}$ representing users, groups of them, or transactions operating on behalf of users. *Access types (privileges)* are the set of database operations such as select, insert, delete, update, execute, grant, or revoke and predicate $p \in P$ defines the access window of subject $s \in S$ on object $o \in O$. The tuple $\langle o, s, t, p \rangle$ is called *access rule* and a function f is defined to determine if an authorization $f(o, s, t, p)$ is valid or not: $f: O \times S \times T \times P \rightarrow \{\text{True}, \text{False}\}$.

For any $\langle o, s, t, p \rangle$, if $f(o, s, t, p)$ evaluates into True, subject s has authorization t to access object o within the range defined by predicate p .

An important property of discretionary security models is the support of the principle of *delegation of rights* where a right is the (o, t, p) -portion of the access rule. A subject s_i who holds the right (o, t, p) may be allowed to delegate that right to another subject s_j ($i \neq j$).

Though most commercial DBMS products still support discretionary protection, it cannot be recommended as access control technique for databases with high security needs. Principally, this is because of the support of delegation of rights and a missing information flow control policy. Moreover, the users of the database and not a central

1. However, audit records are often stored and examined by using the DBMS software.

only capable to support DAC the relational schemas are candidates for implementation and the view descriptors are used for discretionary access controls. In the case the DBMS under consideration supports MAC further design activities are necessary. The Requirements Analysis, Conceptual and Logical Design phases in AMAC are described in [10], for a deeper study of the later phases of AMAC design we refer to [11].

(3) *The AMAC security object.* In order to enforce mandatory security it is necessary to determine security objects and security subjects which are both subject to security label assignments. In AMAC a security object is a database *fragment* and a subject is a *view*. Fragments are derived by using structured database decomposition and views are derived by combining these fragments. A fragment is the largest area of the database to which two or more views have access in common. Additionally, no view exists that has access to a subset of the fragment only. We have developed the structured decomposition approach to derive fragments and our work includes techniques and algorithms for a lossless database decomposition into fragments and algorithms to keep fragmented databases consistent during database update.

(4) *Support of automated security labeling.* As in most IT applications labeled data is not available, AMAC offers a supporting policy for the automated security labeling of security objects and security subjects. Automated labeling is based on the following assumption: The larger the number of users cleared to access a particular fragment, the lower is the sensitivity of the contained data and as a consequence the lower is the level of classification that needs to be provided for the fragment. This assumption seems to be valid because a fragment that is accessed by many users will not contain sensitive information and on the other hand, a fragment that is accessible for few users only can be classified as being highly sensitive. Views (respectively the users having the view as their access window to the data) are ordered based on the number of fragments they may access (they are defined over) and additionally based on the assigned classifications for the fragments. In general, a view needs a clearance that allows the corresponding users to access all fragments the view is defined over. The suggested classification $class(F)$ applies

to the whole fragmental schema F as well as to all attribute names and type definitions for the schema while the suggested clearance $clear(V)$ to all transactions executing on behalf of a user V . It should be noted that classifications and clearances are only candidates for security labels and may be refined by a human database designer if necessary.

(5) *Security Enforcement*. In AMAC the fragments are physically stored and access to a fragment may be controlled by a reference monitor. In AMAC security constraints are handled during database design as well as during query processing. During database design they are expressed by the database decomposition while during query processing they are enforced by *trigger mechanisms*. Triggers are hidden rules that can be fired (activated) if a fragment is effected by certain database operations. In databases security critical operations are the select (read access), the insert, delete, and update (write accesses) commands. In AMAC *select-triggers* are used to route queries to the proper fragments, *insert-triggers* are responsible to decompose tuples and to insert corresponding sub-tuples into proper fragments, and *update-* and *delete-triggers* are responsible for protecting against unauthorized modification by restricting information flow from *high* to *low* in cases that could lead to an undesired information transfer. The operational semantics of the AMAC database operations and the construction of the *select-* and *insert-triggers* are outlined in [12].

A few final words on database security policies are in order. Although the models discussed differ significantly we believe all of the approaches have their right to exist. The discretionary security approach may be the first choice if a high degree of security is not necessary. Keeping the responsibility to enforce security at the users side is only adequate if potential threats against security will not result into considerable damage. Even in the case a central authority is responsible for granting and revoking authorizations DAC-based protection may still be subject to Trojan Horse attacks and cannot be recommended as a security technique for security critical database applications. Mandatory policies are more effective because they imply that users do not have control over the creation and alteration of security parameters. A security policy suitable for a particular application

authority are responsible for enforcing a security policy of an organization. The 'delegation of rights' offers a serious potential for Trojan Horse attacks and revocations of previously granted privileges may result in cascading revocation chains in the authorization graph. Thus, revocation of privileges cannot always be guaranteed! The most simple attack discretionary protection systems cannot withstand is the situation in which an authorized user simply copies the information into a relation with different authorizations.

2.2 Mandatory Access Controls

In addition to access controls mandatory protection systems are concerned with the flow of information within a system as well. Mandatory security requires that security objects and subjects are assigned to certain security levels represented by a label. The label for an object o is called its *classification* ($class(o)$) and a label for a subject s is called its *clearance* ($clear(s)$). The classification represents the sensitivity of the labeled data while the clearance of a subject its trustworthiness to do not disclose sensitive information to others. Mandatory access control (MAC) requirements are often stated based on Bell and LaPadula [4] and are formalized by two rules. The first (simple property) protects the information of the database from unauthorized disclosure, the second (*-property) protects data from contamination or unauthorized modification by restricting the information flow from high to low.

(1) Subject s is allowed to read data item d if $clear(s) \geq class(d)$.

(2) Subject s is allowed to write data item d if $clear(s) \leq class(d)$.

Few final sentences on MAC policies are in order. In many discussions confusion has arisen about the fact that in mandatory systems it is not only sufficient to have strong controls over who can read which data. Why is it necessary to include strong controls over who can write which data in systems with high security requirements? The reason is that a

system with high security needs must protect itself against attacks from unauthorized as well as from authorized users. There are several ways authorized users may disclose sensitive information to others. This can be done by mistake, as a deliberate illegal action, or the user may be tricked to do so by a Trojan horse attack. Consider a simple situation as described above in which an authorized user is disclosing information by retrieving it from the database, copying it into an 'owned' object, and making the copy available to others. To prevent from doing so, it is necessary to control the ability of the authorized user to make a copy (which implies the writing of data). In particular, once a transaction has successfully completed a read attempt, the protection system must ensure that there is no write to a lower security level (*write-down*) that is caused by a user authorized to execute a read transaction. As the read and write checks are both mandatory controls, a MAC system successfully protects against the attempt to copy information and to grant the copy to unauthorized users. By not allowing higher classified subjects to 'write-down' on lower classified data, downward information flow among subjects with different clearances can efficiently be controlled. As covert storage channels require writing to objects, the *-property also helps to limit leakage of information through these hidden paths.

As an interesting optional feature mandatory security and the Bell-LaPadula (BLP) paradigm may lead to multilevel databases. These are databases containing relations which may appear different to users with different clearances. This is due to the following two reasons: Firstly, not all clearances may authorize all subjects to all data and secondly, because of the avoidance of a covert channel the support of MAC may lead to polyinstantiation [5] of attributes or tuples. The basic property in databases is that one fact of reality may only be represented in the database once. Because of polyinstantiation this basic property is no longer valid and this opens a wide range of research on multilevel secure (MLS) data models. For recent proposals consult [6] or [7].

At present MAC in DBMSs as a protection system is supported by several implementations including academic experimental prototype systems (for example,

may also have both a mandatory and a discretionary component. Although very effective mandatory policies can only be applied in environments where labeled information is available. We believe that this is one of the strongest points in favor of the AMAC security model. AMAC includes discretionary as well as mandatory aspects and additionally offers a design environment with main emphasis given on the security of the resulting database.

3. Future Research Directions in Database Security

The field of database security has been very active in the recent years and interest has been dramatically increased in the past few years. This interest is not only reflected by the amount of people interested in the field or by the number of high-quality papers submitted to database- and security conferences but also by national and international computer security centers and standardization organizations working on standards and evaluation criteria or by the International Federation of Information Processing Societies (IFIP) forming a working group (IFIP WG 11.3) on Database Security in 1987. As already has been mentioned in the introduction to this paper during the early stages of research focus was mainly directed towards the discretionary aspect of database security. Later, the focus has shifted to mandatory controls, integrity issues, and security mechanisms fine-tuned to provide privacy. The following contains our interpretation of some directions in which we expect the entire field to move within the next few years.

System Architecture of Mandatory Systems

Most DBMSs supporting MAC are based on the following architecture: The DBMS is hosted on a TCB (trusted computing base) which is responsible for identification, user authentication, and mandatory access controls. Multilevel relations are only supported at an external level and the whole database is decomposed into single-level fragments which are stored using the storage manager of a general purpose DBMS product. In our opinion this approach may have several practical advantages but represents only a near-term

solution to database security. What is necessary in the near future is to develop data models, storage structures, and transaction and recovery management procedures specially suited for the use in DBMSs with a high degree of trust in their security features. A first step into this direction has already been performed for secure transaction management (for example, see [13], [14]) and recovery management (for example, see [15]).

Formal Specification and Verification of MLS DBMSs

Assurance that the security features of a DBMS are working as advertised is required for DBMSs holding databases with security critical content. This involves a formal specification and verification of the DBMS specifications, of the DBMS architecture, of the DBMS implementation, as well as of the design and implementation of a particular database application. So far, not much work has been done on this topic and there is only very little experience using existing systems and techniques to formally specify and verify databases. A natural next step should be to adopt existing techniques and to use them for designing and implementing secure databases. A very good discussion on pros and cons of formal methods within the framework of safety-critical systems has been made in [16].

Evaluation Criteria

It is commonly agreed that the evaluation criteria are a first step into the right direction. However, since the international field of information technology providers will not be able to evaluate their products against different criteria and standards in different countries the criteria proposed will have to be merged. This also includes the need for mutual recognition of security certifications and evaluations between different countries. Moreover, as technology evolves, the concept of security will need to be extended to an open, heterogeneous, multi-vendor environment. In the future solutions different from the systems used today will have to be considered for evaluation, for example, object-oriented systems, knowledge-based systems, active systems, multi-media systems, or hypertext systems may become candidates for evaluation. To cover future development, criteria must be open ended and address the needs of new not yet explored information technology

SeaView from SRI, LDV from MITRE and ASD_Views from TRW) and commercial products from Informix, Ingres, Oracle, Sybase, and others. The systems were built in order to get evaluated at the higher classes of security of the evaluation criteria. For a comparison and study of the systems consult [8].

Though being more restrictive than DAC models it is the opinion of the authors that pure MAC techniques need some extensions in order to be successfully applied to databases. In particular, we see as limitations the following drawbacks in MLS databases supporting mandatory access controls based on BLP: First, there is still no agreement about the answer to the question what should be the granularity of labeled data. Proposals range from protecting whole databases, protecting files, protecting relations, attributes, or even certain attribute values. In any case, careful labeling is necessary because otherwise it could lead to inconsistent or incomplete label assignments. Second, as databases usually contain a large collection of data, serve many users, and labeled data is not available in many civil applications, manual security labeling of data is necessary. However, for large databases this may result in an almost endless process. Third, because of strict information flow policies higher cleared users are in any case restricted from writing-down on lower classified data items. However, organizational policies may require that certain tasks need to be carried out by two or more persons (for example, the four-eyes-principle as common in banking organizations) having different clearances. Following Bell-LaPadula's write-access rule would require the same level of clearance for all parties involved.

2.3 The Adapted Mandatory Access Control Model

The Adapted Mandatory Access Control (AMAC) model goes beyond the approaches described above in two respects. First, it overcomes the drawbacks of the pure MAC approach because it has been developed with the limitations in mind. Second, AMAC does not only cover access control issues and information flow controls but includes in addition a database design environment with main emphasis on the security of resulting

databases. Resulting databases may be implemented in DBMSs supporting DAC only or supporting DAC and MAC. The technique combines well known and widely accepted concepts from the field of data modelling with concepts from the area of data security research. By using AMAC the following design phases for security critical databases can be identified:

(1) *Requirements Analysis and Conceptual Design.* Based on the role they perform in the organization the potential users of the database can be classified into different groups. For different roles data and security requirements may differ significantly. The Entity-Relationship (ER) model [9] and its variants serve as an almost de facto standard for conceptual database design and have been extended in AMAC to model and describe security requirements. The security and data requirements of each role performed in the organization are described by individual ER-schemas and form the view (perception) of each user group on the enterprise data. Please note, in this setting the notion of a view denotes all the information a user performing a certain role in the organization is aware of. This information includes data, security requirements, and functions. Thus, the notion of views appears different from that in a DAC environment where views are unmaterialized queries and can be used for authorization purposes. In order to arrive at a conceptualization of the whole information system as seen from the viewpoint of the enterprise AMAC uses schema integration techniques in a further design step. The resulting conceptual database model is described by a single ER-schema extended by security flags indicating security requirements for certain user roles.

(2) *Logical Design.* In order to implement the conceptual schema into a DBMS a transformation from the ER-schema into the data model supported by the DBMS in use is necessary. AMAC contains general rules and guidelines for the translation of ER-schemas into the relational data model. Output of the transformation process is a set of *relational schemas*, *global dependencies* defined between schemas and necessary for database consistency during further design steps, and *a set of views*, now describing access requirements on relation schemas. If the DBMS that should hold the resulting database is

environments.

Extending Security to Non-relational Data Models

It is only recently that security is discussed in the context of non-relational data models. Preliminary work has begun to develop security models for object-oriented databases (for example, for multilevel approaches see [17], [18], [19], or [20], or for discretionary models, for example, [21], [22], [23], or [24]), for knowledge-based systems, for multimedia databases and for hypertext systems [25]. All the approaches have adopted existing data models for the use in security critical environments. It is expected that further research will lead to new data models in which security will be one of the major initial design decisions.

Research Issues in Discretionary Security

The presence of more advanced data models, e. g. the object-oriented data model, has renewed interest in discretionary access controls. Further research issues include explicit negative authorization, group authorization, propagation of authorization, propagation of revocations, authorizations on methods and functions, and the support of roles.

Design Aids and Tools

Future research is necessary to develop aids and tools supporting a designer during the different phases involved in the design of a database with security critical content. Research is needed in an integrated fashion and must span requirements analysis, conceptual- and logical design, security semantics, integrity rules, and also prototyping, testing, and benchmarking. Aids, guidelines and tools are needed for both discretionary and mandatory protected databases. The AMAC security technique has already made a considerable contribution to this topic. However, more work, in particular work on design aids for the pure mandatory approach is necessary. In a project at the UoV we are working on these challenging issues. For first results, see [26] and [27].

Extending Security to Distributed and Heterogeneous Databases

Distribution adds a further dimension to security because distributed systems are vulnerable to a number of additional security attacks, for example, to data communication attacks. Even more complicated is the case wherein heterogeneous DBMSs are chosen to form a database federation. Since the participating component databases remain operating autonomously and the security mechanisms may be different across the sites, additional security gateways and controls may be necessary. The steps involved in building a secure distributed heterogeneous DBMS are by no means straightforward and some researchers believe that given the current state-of-the-art of both database security and federated database technology, they are not even feasible.

Security and Privacy

Addressing security and privacy themes must remain a future topic of database research. Security and privacy is one of the most important topics in medical informatics, for example, in integrated hospital information systems. In numerous medical venues computerized information systems have been introduced with little regard to security and privacy controls so far. It is a future challenge to database security to cope with the availability, confidentiality, and privacy of computer-based patient records in the near future. An important contribution to this topic has been made by the results of the DORIS project ([28], [29]) but to give more attention to privacy issues will be still necessary in the future.

4. Conclusion

It has been already said that there is a growing interest in database security. The approaches reported in the literature show that there has been considerable success in developing solutions to the problems involved. However, as we have shown above the security of database still remains one of the major and fascinating research areas of the

day. It is expected that changing technology will introduce new vulnerabilities to database security. Together with problems not yet adequately solved database security promises to remain an important area of future research

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